Example 5.34

The parabolic communication antenna has a focus at $2m$ distance from the vertex of the antenna. Find the width of the antenna $3m$ from the vertex.

Solution

Let the parabola be $y^2 = 4ax$.

Since focus is $2m$ from the vertex $a = 2$

Equation of the parabola is $y^2 = 8x$

Let P be a point on the parabola whose x-coordinate is $3m$ from the vertex $P(3, y)$

$$y^2 = 8 \times 3$$
$$y = \sqrt{8 \times 3}$$
$$= 2\sqrt{6}$$

The width of the antenna $3m$ from the vertex is $4\sqrt{6}\, m$.

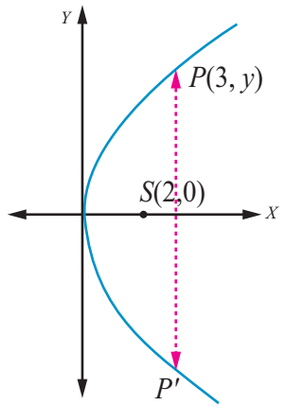


Fig. 5.58

5.7.4 Reflective property of parabola

The light or sound or radio waves originating at a parabola's focus are reflected parallel to the parabola's axis (Fig. 5.60) and conversely the rays arriving parallel to the axis are directed towards the focus (Fig. 5.59).

Example 5.35

The equation $y = \frac{1}{32}x^2$ models cross sections of parabolic mirrors that are used for solar energy. There is a heating tube located at the focus of each parabola; how high is this tube located above the vertex of the parabola?

Solution

Equation of the parabola is

$$y = \frac{1}{32}x^2$$

That is $x^2 = 32y$; the vertex is $(0,0)$

$$= 4(8)y$$

$$\Rightarrow a = 8$$

So the heating tube needs to be placed at focus $(0, a)$. Hence the heating tube needs to be placed 8 units above the vertex of the parabola.

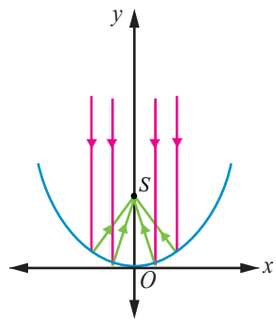


Fig. 5.59

Example 5.36

A search light has a parabolic reflector (has a cross section that forms a 'bowl'). The parabolic bowl is $40\,cm$ wide from rim to rim and $30\,cm$ deep. The bulb is located at the focus .

(1) What is the equation of the parabola used for reflector?

(2) How far from the vertex is the bulb to be placed so that the maximum distance covered?

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